SCHOOL OF MATHEMATICAL SCIENCES

B SC (HONS) IN ACTUARIAL STUDIES
B SC (HONS) IN INDUSTRIAL STATISTICS
BACHELOR OF INFORMATION SYSTEMS (HONS) (DATA ANALYTICS)

ACADEMIC SESSION: SEPTEMBER 2023 SEMESTER

MAT1024 LINEAR ALGEBRA & APPLICATIONS

ASSIGNMENT

DUE DATE: 22 DECEMBER 2023, 12 NOON

Group Member

No.	**Name**	**Student ID**
1	Ooi Jin Yon	22039069
2	Oon Jing Xuan	22042063
3	Nuhaa Khaleda Binti Mohd Khalid	22086508
4	Lee Hong Yen	21090782
5	Christopher Lim Ge Yong	22049472
6	Ngoo Qi Peng	13080775

Group Number	**25**

According to the instructions, we must use the following values for A, B, C, D, E and F to solve the questions:

> *A* = **LAST TWO** digits of **FIRST** group member;
> *B* = **LAST TWO** digits of **SECOND** group member;
> *C* = **LAST TWO** digits of **THIRD** group member;
> *D* = **LAST TWO** digits of **FOURTH** group member;
> *E* = **LAST TWO** digits of **FIFTH** group member;
> *F* = **LAST TWO** digits of **SIXTH** group member.

Digit	Group Member	Number for digit
A	Ooi Jin Yon	69
B	Oon Jing Xuan	63
C	Nuhaa Khaleda Binti Mohd Khalid	08
D	Lee Hong Yen	82
E	Christopher Lim Ge Yong	72
F	Ngoo Qi Peng	75

Question 1

Variables:

X: Number of ounces of Food 1

Y: Number of ounces of Food 2

Z: Number of ounces of Food 3

$2x + 3y + z = A$

$2x + y + 3z = B$

$4x + 2y + 5z = C$

$$\begin{pmatrix} 2 & 3 & 1 \\ 2 & 1 & 3 \\ 4 & 2 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} A \\ B \\ C \end{pmatrix}$$

$$\begin{pmatrix} 2 & 3 & 1 \\ 2 & 1 & 3 \\ 4 & 2 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 69 \\ 63 \\ 8 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \left(\begin{array}{ccc|c} 2 & 3 & 1 & 69 \\ 2 & 1 & 3 & 63 \\ 4 & 2 & 5 & 8 \end{array}\right)$$

Let perform Gaussian Elimination (REF) and Backward Substitution to get the final value of x, y, and z.

$$\left(\begin{array}{ccc|c} 2 & 3 & 1 & 69 \\ 2 & 1 & 3 & 63 \\ 4 & 2 & 5 & 8 \end{array}\right) \xrightarrow{R_1 \to \left(\frac{1}{2}\right)R_1} \left(\begin{array}{ccc|c} 1 & \frac{3}{2} & \frac{1}{2} & \frac{69}{2} \\ 2 & 1 & 3 & 63 \\ 4 & 2 & 5 & 8 \end{array}\right) \xrightarrow{R_2 \to R_2 + (-2)R_1} \left(\begin{array}{ccc|c} 1 & \frac{3}{2} & \frac{1}{2} & \frac{69}{2} \\ 0 & -2 & 2 & -6 \\ 4 & 2 & 5 & 8 \end{array}\right)$$

$$\xrightarrow{R_3 \to R_3 + (-4)R_1} \left(\begin{array}{ccc|c} 1 & \frac{3}{2} & \frac{1}{2} & \frac{69}{2} \\ 0 & -2 & 2 & -6 \\ 0 & -4 & 3 & -130 \end{array}\right) \xrightarrow{R_2 \to \left(-\frac{1}{2}\right)R_2} \left(\begin{array}{ccc|c} 1 & \frac{3}{2} & \frac{1}{2} & \frac{69}{2} \\ 0 & 1 & -1 & 3 \\ 0 & -4 & 3 & -130 \end{array}\right)$$

$$\xrightarrow{R_3 \to R_3 + (4)R_2} \left(\begin{array}{ccc|c} 1 & \frac{3}{2} & \frac{1}{2} & \frac{69}{2} \\ 0 & 1 & -1 & 3 \\ 0 & 0 & -1 & -118 \end{array}\right) \xrightarrow{R_3 \to (-1)R_3} \left(\begin{array}{ccc|c} 1 & \frac{3}{2} & \frac{1}{2} & \frac{69}{2} \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & 118 \end{array}\right)$$

Backward Substitution:

$$\left(\begin{array}{ccc|c} 1 & \frac{3}{2} & \frac{1}{2} & \frac{69}{2} \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & 118 \end{array}\right)$$

$$x + \frac{3}{2}y + \frac{1}{2}z = \frac{69}{2}$$

$$x + \frac{3}{2}(121) + \frac{1}{2}(118) = \frac{69}{2}$$

$$x + \frac{363}{2} + 59 = \frac{69}{2}$$

$$x = \frac{69}{2} - 59 - \frac{363}{2}$$

$$x = -206$$

$$y - z = 3$$

$$y - 118 = 3$$

$$y = 3 + 118$$

$$y = 121$$

$$z = 118$$

Let perform Gauss-Jordan elimination (or RREF) to get the final value of x, y, and z

$$\left(\begin{array}{ccc|c} 2 & 3 & 1 & 69 \\ 2 & 1 & 3 & 63 \\ 4 & 2 & 5 & 8 \end{array}\right) \xrightarrow{R_1 \to \left(\frac{1}{2}\right)R_1} \left(\begin{array}{ccc|c} 1 & \frac{3}{2} & \frac{1}{2} & \frac{69}{2} \\ 2 & 1 & 3 & 63 \\ 4 & 2 & 5 & 8 \end{array}\right) \xrightarrow{R_2 \to R_2 + (-2)R_1} \left(\begin{array}{ccc|c} 1 & \frac{3}{2} & \frac{1}{2} & \frac{69}{2} \\ 0 & -2 & 2 & -6 \\ 4 & 2 & 5 & 8 \end{array}\right)$$

$$\xrightarrow{R_3 \to R_3 + (-4)R_1} \left(\begin{array}{ccc|c} 1 & \frac{3}{2} & \frac{1}{2} & \frac{69}{2} \\ 0 & -2 & 2 & -6 \\ 0 & -4 & 3 & -130 \end{array}\right) \xrightarrow{R_2 \to \left(-\frac{1}{2}\right)R_2} \left(\begin{array}{ccc|c} 1 & \frac{3}{2} & \frac{1}{2} & \frac{69}{2} \\ 0 & 1 & -1 & 3 \\ 0 & -4 & 3 & -130 \end{array}\right)$$

$$\xrightarrow{R_3 \to R_3 + (4)R_2} \left(\begin{array}{ccc|c} 1 & \frac{3}{2} & \frac{1}{2} & \frac{69}{2} \\ 0 & 1 & -1 & 3 \\ 0 & 0 & -1 & -118 \end{array}\right) \xrightarrow{R_3 \to (-1)R_3} \left(\begin{array}{ccc|c} 1 & \frac{3}{2} & \frac{1}{2} & \frac{69}{2} \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & 118 \end{array}\right)$$

$$\xrightarrow{R_1 \to R_1 + \left(-\frac{3}{2}\right)R_2} \left(\begin{array}{ccc|c} 1 & 0 & 2 & 30 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & 118 \end{array}\right) \xrightarrow{R_1 \to R_1 + (-2)R_3} \left(\begin{array}{ccc|c} 1 & 0 & 0 & -206 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & 118 \end{array}\right)$$

$$\xrightarrow{R_2 \to R_2 + R_3} \left(\begin{array}{ccc|c} 1 & 0 & 0 & -206 \\ 0 & 1 & 0 & 121 \\ 0 & 0 & 1 & 118 \end{array}\right)$$

$x = -206\ ounces$

$y = 121\ ounces$

$Z = 118\ ounces$

Question 2

(a)

Answer:

$$Transition\ matrix = \begin{pmatrix} 0.82 & 0.28 \\ 0.18 & 0.72 \end{pmatrix}$$

Explanation:

The matrix above is the transition matrix for this phenomenon

Air Quality	Good	Bad
Good	82% or 0.82 (D)	28% or 0.28
Bad	18% or 0.18	72% or 0.72 (E)
Total (Probability = 1)	1	1

(b)

$$\underset{\sim}{x}^{(1)} = P\underset{\sim}{x}^{(0)} = \begin{pmatrix} 0.82 & 0.28 \\ 0.18 & 0.72 \end{pmatrix} \times \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0.82 \\ 0.18 \end{pmatrix}$$

$$\underset{\sim}{x}^{(2)} = P\underset{\sim}{x}^{(1)} = \begin{pmatrix} 0.82 & 0.28 \\ 0.18 & 0.72 \end{pmatrix} \times \begin{pmatrix} 0.82 \\ 0.18 \end{pmatrix}$$

$$= \begin{pmatrix} 0.7228 \\ 0.2772 \end{pmatrix}$$

$0.7228 \times 100 = 72.28\%$

Answer: *There is 72.28% chances that the air quality will be good two days from now.*

(c)

$$\underset{\sim}{x}^{(1)} = P\underset{\sim}{x}^{(0)} = \begin{pmatrix} 0.82 & 0.28 \\ 0.18 & 0.72 \end{pmatrix} \times \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0.28 \\ 0.72 \end{pmatrix}$$

$$\underset{\sim}{x}^{(2)} = P\underset{\sim}{x}^{(1)} = \begin{pmatrix} 0.82 & 0.28 \\ 0.18 & 0.72 \end{pmatrix} \times \begin{pmatrix} 0.28 \\ 0.72 \end{pmatrix}$$

$$= \begin{pmatrix} 0.4312 \\ 0.5688 \end{pmatrix}$$

$$\underset{\sim}{x}^{(3)} = P\underset{\sim}{x}^{(2)} = \begin{pmatrix} 0.82 & 0.28 \\ 0.18 & 0.72 \end{pmatrix} \times \begin{pmatrix} 0.4312 \\ 0.5688 \end{pmatrix}$$

$$= \begin{pmatrix} 0.5128 \\ 0.4872 \end{pmatrix}$$

$$0.4872 \times 100 = 48.72\%$$

Answer: *There is* 48.72% *chance that the air quality will be bad three days from now.*

(d)

$$P\begin{pmatrix} 0.2 \\ 0.8 \end{pmatrix} = \begin{pmatrix} 0.82 & 0.28 \\ 0.18 & 0.72 \end{pmatrix} \times \begin{pmatrix} 0.2 \\ 0.8 \end{pmatrix}$$

$$= \begin{pmatrix} 0.388 \\ 0.612 \end{pmatrix}$$

Answer: *There will be* 38.8% *chance that air quality will be good tomorrow.*

Question 3

(a)

$$E = \begin{pmatrix} 0.1 & 0.6 & 0.4 \\ 0.3 & 0.2 & 0.3 \\ 0.4 & 0.1 & 0.2 \end{pmatrix}$$

(b)

D=82, E=72, F=75

The linear system $(I - C)x = d$ is then

$$\begin{pmatrix} 0.9 & -0.6 & -0.4 \\ -0.3 & 0.8 & -0.3 \\ -0.4 & -0.1 & 0.8 \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix} = \begin{pmatrix} D_1 \\ D_2 \\ D_3 \end{pmatrix} = \begin{pmatrix} 82 \\ 72 \\ 75 \end{pmatrix}$$

The solution is given by $x = (I - C)^{-1}d$

$$(I - C)^{-1} = \frac{1}{|(I - C)|} adj\,(I - C)$$

$$Det \begin{pmatrix} 0.9 & -0.6 & -0.4 \\ -0.3 & 0.8 & -0.3 \\ -0.4 & -0.1 & 0.8 \end{pmatrix} = 0.9 \begin{vmatrix} 0.8 & -0.3 \\ -0.1 & 0.8 \end{vmatrix} + 0.6 \begin{vmatrix} -0.3 & -0.3 \\ -0.4 & 0.8 \end{vmatrix} - 0.4 \begin{vmatrix} -0.3 & 0.8 \\ -0.4 & -0.1 \end{vmatrix}$$

$$= 0.9(0.64 - 0.03) + 0.6(-0.24 - 0.12) - 0.4(0.03 + 0.32)$$

$$= 0.549 - 0.216 - 0.14$$

$$= 0.193$$

Adjoint:

$M_{11} = (0.8)(0.8) - (-0.3)(-0.1)$ $= 0.61$ $C_{11} = 0.61$	$M_{12} = (-0.3)(0.8) - (-0.3)(-0.4)$ $= -0.36$ $C_{12} = 0.36$	$M_{13} = (-0.3)(-0.1) - (0.8)(-0.4)$ $= 0.35$ $C_{13} = 0.35$
$M_{21} = (-0.6)(0.8) - (-0.4)(-0.1)$ $= -0.52$ $C_{21} = 0.52$	$M_{22} = (0.9)(0.8) - (-0.4)(-0.4)$ $= 0.56$ $C_{22} = 0.56$	$M_{23} = (0.9)(-0.1) - (-0.6)(-0.4)$ $= -0.33$ $C_{23} = 0.33$

$M_{31} = (-0.6)(-0.3) - (-0.4)(0.8)$ $= 0.50$ $C_{31} = 0.50$	$M_{32} = (0.9)(-0.3) - (-0.4)(-0.3)$ $= -0.39$ $C_{32} = 0.39$	$M_{33} = (0.9)(0.8) - (-0.6)(-0.3)$ $= 0.54$ $C_{33} = 0.54$

$$\text{Adj(I-C)}= \begin{pmatrix} C11 & C12 & C13 \\ C21 & C22 & C23 \\ C31 & C32 & C33 \end{pmatrix}^T = \begin{pmatrix} 0.61 & 0.36 & 0.35 \\ 0.52 & 0.56 & 0.33 \\ 0.50 & 0.39 & 0.54 \end{pmatrix}^T = \begin{pmatrix} 0.61 & 0.52 & 0.50 \\ 0.36 & 0.56 & 0.39 \\ 0.35 & 0.33 & 0.54 \end{pmatrix}$$

$$A^{-1} = \frac{1}{0.193}\begin{pmatrix} 0.61 & 0.52 & 0.50 \\ 0.36 & 0.56 & 0.39 \\ 0.35 & 0.33 & 0.54 \end{pmatrix}$$

$$= \frac{1}{193}\begin{pmatrix} 610 & 520 & 500 \\ 360 & 560 & 390 \\ 350 & 330 & 540 \end{pmatrix}$$

$$\underset{\sim}{x} = (I - \widetilde{C})^{-1}\underset{\sim}{d}$$

$$\underset{\sim}{x} = \frac{1}{193}\begin{pmatrix} 610 & 520 & 500 \\ 360 & 560 & 390 \\ 350 & 330 & 540 \end{pmatrix}\begin{pmatrix} 82 \\ 72 \\ 75 \end{pmatrix}$$

$$= \begin{pmatrix} 647.4611399 \\ 513.4196891 \\ 481.6580311 \end{pmatrix}$$

$$= \begin{pmatrix} 647.46 \\ 513.42 \\ 481.66 \end{pmatrix}$$

Answer:

The production vector is $\begin{pmatrix} 647.46 \\ 513.42 \\ 481.66 \end{pmatrix}$

The total output of the housing should be $647.46, the total output of the food should be $513.42, the total output of the utilities should be $481.66.

Question 4

(a)

According to the table, A is 69, C is 8, E is 72 and F is 75.

$$\begin{pmatrix} A & C \\ E & F \end{pmatrix} = \begin{pmatrix} 69 & 8 \\ 72 & 75 \end{pmatrix}$$

Group the plaintext into pair of letters, WE, LO, VE, MA, TH, SS. The dummy letter S is introduced to fill out the last pair.

$$\text{WE} = \begin{pmatrix} 69 & 8 \\ 72 & 75 \end{pmatrix}\begin{pmatrix} 23 \\ 5 \end{pmatrix} = \begin{pmatrix} 1627 \\ 2031 \end{pmatrix} \; or \; \begin{pmatrix} 15 \\ 3 \end{pmatrix} (mod\ 26) = OC$$

$$\text{LO} = \begin{pmatrix} 69 & 8 \\ 72 & 75 \end{pmatrix}\begin{pmatrix} 12 \\ 15 \end{pmatrix} = \begin{pmatrix} 948 \\ 1989 \end{pmatrix} \; or \; \begin{pmatrix} 12 \\ 13 \end{pmatrix} (mod\ 26) = LM$$

$$\text{VE} = \begin{pmatrix} 69 & 8 \\ 72 & 75 \end{pmatrix}\begin{pmatrix} 22 \\ 5 \end{pmatrix} = \begin{pmatrix} 1558 \\ 1959 \end{pmatrix} \; or \; \begin{pmatrix} 24 \\ 9 \end{pmatrix} (mod\ 26) = XI$$

$$\text{MA} = \begin{pmatrix} 69 & 8 \\ 72 & 75 \end{pmatrix}\begin{pmatrix} 13 \\ 1 \end{pmatrix} = \begin{pmatrix} 905 \\ 1011 \end{pmatrix} \; or \; \begin{pmatrix} 21 \\ 23 \end{pmatrix} (mod\ 26) = UW$$

$$\text{TH} = \begin{pmatrix} 69 & 8 \\ 72 & 75 \end{pmatrix}\begin{pmatrix} 20 \\ 8 \end{pmatrix} = \begin{pmatrix} 1444 \\ 2040 \end{pmatrix} \; or \; \begin{pmatrix} 14 \\ 12 \end{pmatrix} (mod\ 26) = NL$$

$$\text{SS} = \begin{pmatrix} 69 & 8 \\ 72 & 75 \end{pmatrix}\begin{pmatrix} 19 \\ 19 \end{pmatrix} = \begin{pmatrix} 1463 \\ 2793 \end{pmatrix} \; or \; \begin{pmatrix} 7 \\ 11 \end{pmatrix} (mod\ 26) = GK$$

Answer: The ciphertext is **OCLMXIUWNLGK.**

(b)

Given that **cipher text is "HUK EST COM"**, and the **plaintext known as "SUNWAY UNI"**

$$Plaintext: \begin{pmatrix} S \\ U \\ N \end{pmatrix} = \begin{pmatrix} 19 \\ 21 \\ 14 \end{pmatrix}, \begin{pmatrix} W \\ A \\ Y \end{pmatrix} = \begin{pmatrix} 23 \\ 1 \\ 25 \end{pmatrix}, \begin{pmatrix} U \\ N \\ I \end{pmatrix} = \begin{pmatrix} 21 \\ 14 \\ 9 \end{pmatrix}$$

$$Ciphertext: \begin{pmatrix} H \\ U \\ K \end{pmatrix} = \begin{pmatrix} 8 \\ 21 \\ 11 \end{pmatrix}, \begin{pmatrix} E \\ S \\ T \end{pmatrix} = \begin{pmatrix} 5 \\ 19 \\ 20 \end{pmatrix}, \begin{pmatrix} C \\ O \\ M \end{pmatrix} = \begin{pmatrix} 3 \\ 15 \\ 13 \end{pmatrix}$$

We need to find $\boldsymbol{X}$ and $\boldsymbol{Y}$ by using following formula, where $\boldsymbol{X}$ **is enciphering matrix** and $\boldsymbol{Y}$ **is deciphering matrix**:

Encryption : $Ciphertext = Key \times Plaintext \ \ (Mod26)$

Decryption : $Plaintext = Key^{-1} \times Ciphertext \ \ (Mod26)$

Where $K^{-1} = \frac{1}{|K|} adj\ k$

$$\boldsymbol{Encryption}: \begin{pmatrix} \mathbf{8} & \mathbf{5} & \mathbf{3} \\ \mathbf{21} & \mathbf{19} & \mathbf{15} \\ \mathbf{11} & \mathbf{20} & \mathbf{13} \end{pmatrix} = \boldsymbol{X} \begin{pmatrix} \mathbf{19} & \mathbf{23} & \mathbf{21} \\ \mathbf{21} & \mathbf{1} & \mathbf{14} \\ \mathbf{14} & \mathbf{25} & \mathbf{9} \end{pmatrix}$$

$$Det \begin{pmatrix} 19 & 23 & 21 \\ 21 & 1 & 14 \\ 14 & 25 & 9 \end{pmatrix} = 19 \begin{vmatrix} 1 & 14 \\ 25 & 9 \end{vmatrix} - 23 \begin{vmatrix} 21 & 14 \\ 14 & 9 \end{vmatrix} + 21 \begin{vmatrix} 21 & 1 \\ 14 & 25 \end{vmatrix}$$

$$= 19(9 - 350) - 23(189 - 196) + 21(525 - 14)$$

$$= 19(-341) - 23(-7) + 21(511)$$

$$= -6479 + 161 + 10731$$

$$= 4413$$

$$= 19\ (mod\ 26)$$

Adjoint P = Transpose of Cofactor P

$M_{11} = (1)(9) - (25)(14)$ $= -341$ $C_{11} = -341$	$M_{12} = (21)(9) - (14)(14)$ $= -7$ $C_{12} = 7$	$M_{13} = (21)(25) - (14)(1)$ $= 511$ $C_{13} = 511$
$M_{21} = (23)(9) - (21)(25)$ $= -318$ $C_{21} = 318$	$M_{22} = (19)(9) - (14)(21)$ $= -123$ $C_{22} = -123$	$M_{23} = (19)(25) - (23)(14)$ $= 153$ $C_{23} = -153$
$M_{31} = (23)(14) - (1)(21)$ $= 301$ $C_{31} = 301$	$M_{32} = (19)(14) - (21)(21)$ $= -175$ $C_{32} = 175$	$M_{31} = (19)(1) - (21)(23)$ $= -464$ $C_{31} = -464$

$$\begin{pmatrix} -341 & 7 & 511 \\ 318 & -123 & -153 \\ 301 & 175 & -464 \end{pmatrix}^{T} = \begin{pmatrix} -341 & 318 & 301 \\ 7 & -123 & 175 \\ 511 & -153 & -464 \end{pmatrix}$$

$$K^{-1} = \frac{1}{|K|} adj\ k$$

$$A^{-1} = \frac{1}{4413}\begin{pmatrix} -341 & 318 & 301 \\ 7 & -123 & 175 \\ 511 & -153 & -464 \end{pmatrix}$$

$$= 19^{-1}\begin{pmatrix} -341 & 318 & 301 \\ 7 & -123 & 175 \\ 511 & -153 & -464 \end{pmatrix}$$

$$= 11\begin{pmatrix} -341 & 318 & 301 \\ 7 & -123 & 175 \\ 511 & -153 & -464 \end{pmatrix}$$

$$= \begin{pmatrix} -3751 & 3498 & 3311 \\ 77 & -1353 & 1925 \\ 5621 & -1683 & -5104 \end{pmatrix}$$

$$= \begin{pmatrix} 19 & 14 & 9 \\ 25 & 25 & 1 \\ 5 & 7 & 18 \end{pmatrix} (mod\ 26)$$

$$X = \begin{pmatrix} 8 & 5 & 3 \\ 21 & 19 & 15 \\ 11 & 20 & 13 \end{pmatrix}\begin{pmatrix} 19 & 14 & 9 \\ 25 & 25 & 1 \\ 5 & 7 & 18 \end{pmatrix}$$

$$X = \begin{pmatrix} 292 & 258 & 131 \\ 949 & 874 & 478 \\ 774 & 745 & 353 \end{pmatrix}$$

$$X = \begin{pmatrix} 6 & 24 & 1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{pmatrix} (mod\ 26)$$

X is enciphering matrix

Decryption

$$Det(K) = 6\begin{vmatrix} 16 & 10 \\ 17 & 15 \end{vmatrix} - 24\begin{vmatrix} 13 & 10 \\ 20 & 15 \end{vmatrix} + 1\begin{vmatrix} 13 & 16 \\ 20 & 17 \end{vmatrix}$$

$$= 6(240 - 170) - 24(195 - 200) + (221 - 320)$$

$$= 6(70) - 24(-5) - 99$$

$$= 540 - 99$$

$$= 441$$

$$= 25\ (mod\ 26)$$

$$\frac{1}{\det(X)} = \frac{1}{25}$$

$$= 25^{-1}$$

$$= 25\ (mod\ 26)$$

Adjoint X

$M_{11} = (16)(15) - (10)(17)$ $= 70$ $C_{11} = 70$	$M_{12} = (13)(15) - (10)(20)$ $= 5$ $C_{12} = 5$	$M_{13} = (13)(17) - (16)(20)$ $= -99$ $C_{13} = -99$
$M_{21} = (24)(15) - (1)(17)$ $= 343$ $C_{21} = -343$	$M_{22} = (6)(15) - (1)(20)$ $= 70$ $C_{22} = 70$	$M_{23} = (6)(17) - (24)(20)$ $= -378$ $C_{23} = 378$
$M_{31} = (24)(10) - (1)(16)$ $= 224$ $C_{31} = 224$	$M_{32} = (6)(10) - (1)(13)$ $= 47$ $C_{32} = -47$	$M_{31} = (6)(16) - (24)(13)$ $= -216$ $C_{31} = -216$

$$\begin{pmatrix} 70 & 5 & -99 \\ -343 & 70 & 378 \\ 224 & -47 & -216 \end{pmatrix}^{T} = \begin{pmatrix} 70 & -343 & 224 \\ 5 & 70 & -47 \\ -99 & 378 & -216 \end{pmatrix}$$

$$Y = \frac{1}{25}\begin{pmatrix} 70 & -343 & 224 \\ 5 & 70 & -47 \\ -99 & 378 & -216 \end{pmatrix}$$

$$= 25^{-1}\begin{pmatrix} 70 & -343 & 224 \\ 5 & 70 & -47 \\ -99 & 378 & -216 \end{pmatrix}$$

$$= 25\begin{pmatrix} 70 & -343 & 224 \\ 5 & 70 & -47 \\ -99 & 378 & -216 \end{pmatrix}$$

$$= \begin{pmatrix} 1750 & -8575 & 5600 \\ 125 & 1750 & -1175 \\ -2475 & 9450 & -5400 \end{pmatrix}$$

$$= \begin{pmatrix} 8 & 5 & 10 \\ 21 & 8 & 21 \\ 21 & 12 & 8 \end{pmatrix} (mod\ 26)$$

Y is deciphering matrix

Answer:

$$X = \begin{pmatrix} 6 & 24 & 1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{pmatrix}\ is\ enciphering\ matrix$$

$$Y = \begin{pmatrix} 8 & 5 & 10 \\ 21 & 8 & 21 \\ 21 & 12 & 8 \end{pmatrix}\ is\ deciphering\ matrix$$